AIRCRAFT MAINTENANCE ENGINEER'S LICENCE EXAMINATION
CATEGORY 'C' PISTON ENGINES

1. To comply with Regulations, the Certificate of Fitness for flight under the 'A Condition' must be signed by
   a. a category A (Aircraft) licence holder only
   b. a category A (Aircraft) and a category C (Engineer) licence holders only
   c. all the categories licence holders relevant to the aircraft

2. The radio apparatus required for Nigerian registered aircraft can be found in
   a. the fourth schedule to the Regulations
   b. the fifth schedule to the Regulations
   c. the sixth schedule to the Regulations

3. A defect affecting the engine occurred in flight and was recorded in the technical log by the pilot. The defect was rectified by an engineer. The recording of this defect with its rectification must be made in:
   a. the technical log only
   b. the technical log and the engine log book
   c. the technical log, the engine log book and the aircraft log book

4. Log book entry must be made not later than
   a. seven days after the occurrence
   b. seven days after the expiration of the Certificate of Maintenance (if any) in force in respect of the aircraft at the time of the occurrence
   c. ten days after the occurrence

5. Certificate of maintenance are issued in duplicate:
   a. and shall both be carried in the aircraft
   b. both should be kept on the ground in all cases
   c. one should be kept in the aircraft and the other kept by the operator elsewhere other than in the aircraft

6. A Certificate of Release to Service certifies that
   a. all necessary regulations have been complied with in repairing the aircraft
   b. the part of the aircraft/equipment has been overhauled, repaired, replaced or modified in an approved manner and with approval materials
   c. Rectifications have been carried out on the aircraft

7. Which of the following statements is true:
   a. Any aircraft registered in Nigeria must carry a fireproof metal plate bearing the registration marks together with the name and address of the registered owner of the aircraft
   b. The nameplate is only mandatory for transport category aircraft
   c. Nameplate has nothing to do with the Regulations

8. Maintenance schedule is not mandatory for:
   a. Transport category (Passenger) aircraft
   b. Transport category (Cargo) aircraft
   c. Private category aircraft

9. The technical log is:
   a. Mandatory for all aircraft registered in Nigeria
   b. Mandatory for all aircraft for which maintenance to approved maintenance schedule is mandatory
   c. Mandatory for all Aircraft above 12,500lb

10. For any defect cleared on a technical log:
   a. Signing out the snag by an appropriate Engineer is all that is required
   b. The issue of Certificate of Compliance to be kept in the office after signing out the snag is all that is required
   c. A copy of the Certificate of Compliance shall be entered in the technical log in such a position or manner as to be readily identifiable with entry of the defect to which it relates

11. When a component is replaced:
   a. A Certificate of Compliance is signed
   b. A Certificate of Maintenance is signed
   c. Both Certificate of Compliance and Certificate of Maintenance should be signed

12. Weight schedule shall be preserved until:
   a. Another weighing is carried out
   b. Six months following the next weighing
   c. Two years following the next weighing

13. If an aircraft has four engines with variable pitch propellers, the number of log books to be kept are:
   a. 6
   b. 5
   c. 9

14. After conducting an annual inspection, who is qualified to return the aircraft to service?
   a. A certificated A & C personnel
   b. A certificated mechanic with an inspection authorisation of privileges
   c. An DSRAM inspector only

15. An aircraft is due for an inspection specified in its maintenance schedule, to demand for flying the aircraft the thing to
    a. For a licensed engineer to extend the inspection period and stamp it in the log book before flying the aircraft
    b. To insist that the aircraft must be inspected as specified before it can perform the flight
    c. Fly the aircraft before carrying out the inspection in as much as it has a current Certificate of Airworthiness

16. An engine with a combustion chamber of $0.4m^3$ capacity and a piston displacement of $1.6m^3$ has a compression ratio of:
    a. 5:1
    b. 7:1
    c. 6:1

17. The root mean square of 2, 3, 4 and 5
    a. $\sqrt{(27/2)}$
    b. $\sqrt{54}$
    c. $\sqrt{30}$

18. Given that the bore of an engine's cylinder is 2cm and its stroke is 10cm, then its swept volume is:
    a. $20\pi cm^3$
    b. $5\pi cm^3$
    c. $10\pi cm^3$

19. The cube root of 1/27 is
    a. 3
    b. ½
    c. 1/3

20. The value of 20 to the power of zero i.e. 20° is:
    a. 20
    b. 1
    c. 0

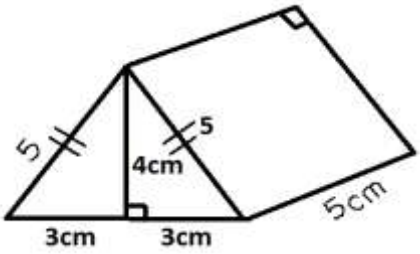
21. On drawing, a dimension is given as $2.50^{+0.001}_{-0.002}$; the tolerance is
    a. 0.002
    b. 0.003
    c. 2.497

22. The principle of a micrometer is that:
    a. the barrel is divided into 50 parts
    b. the ratchet on the thimble makes it easy to retain a reading
    c. the spindle advances 1/40 inch with each revolution of the thimble

23. The total jaw thickness of a vernier caliper is ¼inch; this implies that:
    a. the minimum internal diameter that can be measured is 0.25inch
    b. to obtain the true diameter, 0.25 inch is added to the reading
    c. Both are correct

24. When air is compressed:
    a. the temperature increases and the density decreases
    b. the pressure increases and density decreases
    c. the density increases and temperature increases

25. An aircraft bolt with a single raised star on the head would indicate a:
    a. Corrosion resistant steel bolt
    b. Standard steel bolt
    c. A magnetically inspected bolt

26. 

The figure is a prism with an isosceles triangular section and its dimensions are as shown: What is its volume?
    a. 120cu.cm
    b. 60cu.cm
    c. 75cu.cm

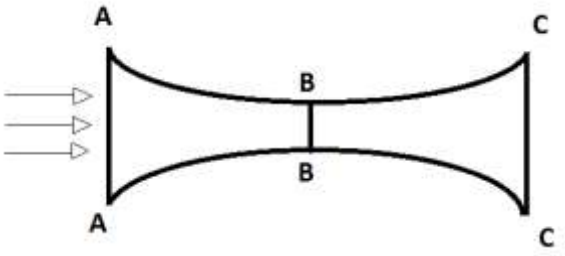
27. If the volume of a gas is constant then:
    a. Its pressure is proportional to its temperature
    b. Its pressure is inversely proportional to its absolute temperature
    c. $P_1V_1 = P_2V_2$

28. Force is proportional to mass and acceleration is:
    a. Newton's 1st law of motion
    b. Newton's 2nd law of motion
    c. Newton's 3rd law of motion

29. The law relating the pressure and absolute temperature of a given mass of gas at constant volume is:
    a. Charles' law
    b. Boyle's law
    c. Newton's law

30. During the combustion cycle of a reciprocating engine, the governing gas law is:
    a. Charles' law
    b. Boyle's law
    c. Bernoulli's Principle

31. Slight corrosion was removed from the fin of a cylinder barrel.
    a. The surface need not to be protected because the original protection was heavy
    b. The surface has to be protected by Enamel coating
    c. The surface has to be protected by priming and coating



32. At section B-B of the above figure, the pressure is:
    a. Maximum
    b. Minimum
    c. The same as in section C-C

33. From the definition of Logarithms, $\log_2 8$ is equal to:
    a. 10
    b. 3
    c. 2

34. The float level of a float-type carburetor is measured in fractions of an inch from:
    a. Bottom of float chamber
    b. Parting line of float bowl
    c. Top of float chamber

35. A carburetor setting should always favour:
    a. A slightly lean mixture
    b. lean-best power mixture
    c. A slightly rich mixture

36. In a fuel system containing the items mentioned in series, what would be the sequence starting from the tank (Fuel pump means engine-driven pump)
    a. Pressure gauge, shut off, hand pump, pressure pump
    b. Hand pump, shut off, fuel pump, pressure gauge
    c. Shut off, hand pump, fuel pump, pressure gauge

37. Extreme humps or bends in fuel lines are to be avoided due to the possibility of
    a. Enriched mixture
    b. Bursting lines
    c. Failure of fuel pump

38. A leak in the induction pipe will result into
    a. A rich mixture
    b. A lean mixture
    c. Mixture remains unaffected

39. Most engine wear on aircraft engines occurs during:
    a. take-off power setting
    b. starting and warm-up period
    c. climb

40. Excessive fluctuation of an oil pressure gauge may be caused by:
    a. Leak in pump housing
    b. Excessive engine vibration
    c. A completely clogged oil pressure gauge line

41. Good design requires that engine oil temperature be taken at:
    a. tank inlet
    b. tank outlet
    c. engine inlet

42. The first thing to check if oil pressure will not build-up sufficiently is:
    a. insufficient oil supply
    b. pressure gauge broken
    c. pressure relief valve setting

43. A scavenge pump:
    a. Supplies pressure oil to the engine
    b. Removes oil from the engine crankcase
    c. Supplies oil to the propeller feathering system

44. Main bearings are sometimes installed with an oil seal at the ends. This seal is primarily to:
    a. Prevent oil leaking into crankcase
    b. Control oil consumption
    c. Prevent oil film forming on cylinder walls

45. Viscosity of oil is an indication of its:
    a. Resistance to being poured
    b. Weight
    c. Reaction to temperature

46. Piston pins are lubricated by:
    a. Oil mist thrown from revolving crankshaft parts
    b. Hand packed grease prior to assembly
    c. Pressure from crankshaft through drilled passage in connecting rod

47. A limit switch is:
    a. A device to limit generator output
    b. A switch to shut off electrical units
    c. A switch to prevent overspeeding of a series motor

48. What is the total resistance in a circuit having an e.m.f. of 24 volts and a current of 3 amps?
    a. 20 ohms
    b. 8 ohms
    c. 72 ohms

49. An instrument to measure the resistance to electrical flow encountered in a conductor through which current is passed is called:
    a. A resistometer
    b. A rheostat
    c. An ohmmeter

50. Aviation fuel, is frequently red, reddish brown, or blue in colour. This colour is caused by:
    a. The presence of tetra-ethyl lead
    b. Dye to indicate the fuel is leaded
    c. Nothing – this is the natural colour of gasoline.

51. It is well known that tetraethyl lead is added to aviation gasoline. Specifically this is for the purpose of:
    a. increasing burning rate
    b. reducing burning rate
    c. increasing the heat value

52. A bourdon-tube of instrument would be associated with what part of fuel system?
    a. Exhaust gas analyzer
    b. Inlet air temperature
    c. Fuel pressure

53. A normal fuel-air ratio will be indicated by which of the following types of exhaust
    a. A long light blue flame with a yellow tip
    b. A short dark blue flame
    c. A purple flame with a blue tip and light smoke

54. An engine lopes and emits black smoke when idling, this may be due to
    a. Clogged idle discharge nozzle
    b. Oversize main metering jet
    c. Oversize idle metering jet

55. Other than preventing metal-to-metal contact of moving metal parts, the lubricating system of an engine is used to:
    a. Aid in dissipation of heat
    b. Prevent inter-crystalline corrosion
    c. Operate ground-adjustable propellers

56. Worn bearings in a four-cylinder opposed aircraft engine would result in:
    a. Low oil consumption
    b. Failure of scavenge pump
    c. Low oil pressure

57. The first thing to check if oil pressure will not build-up sufficiently is:
    a. insufficient oil supply
    b. pressure gauge broken
    c. pressure relief valve setting

58. A scavenge pump:
    a. Supplies pressure oil to the engine
    b. Removes oil from the engine crankcase
    c. Supplies oil to the propeller feathering system

59. A low tension ignition system is:
    a. One having individual coils for each spark plug
    b. A system in which the spark plug voltage is held to only a few hundred volts
    c. Any system designed for use on light aircraft

60. Piston rings are usually made of:
    a. Stainless steel
    b. Grey cast iron
    c. Cast aluminium

61. Nitriding is a process given to cylinder barrels to:
    a. Anneal them
    b. Straighten them
    c. Harden them

62. Aircraft pistons are commonly made of:
    a. Aluminium alloy
    b. Stainless steel
    c. Aluminium bronze

63. Crankshafts are usually constructed of:
    a. Forged nickel steel and heat-treated for hardness
    b. Forged aluminium and heat-treated for hardness
    c. High strength alloy steel and annealed

64. A battery generator system provides direct current. On installations requiring alternating current from the battery-generator system, it is necessary to have:
    a. An inverter
    b. A voltage regulator to provide alternating current
    c. A transformer

65. Conduit is sometimes drilled with small holes in the bottom to:
    a. Provide moisture drain
    b. Give inspection openings
    c. Give static discharge opening

66. The resistance to electrical flow of a wire depends on:
    a. Length and material of wire only
    b. Diameter of wire only
    c. Diameter, length and material of wire

67. The opposition offered by a coil of wire to the flow of alternating current is called:
    a. inductive reactance
    b. conductance resistance
    c. capacitive reactance

68. A reversible propeller is one that:
    a. Will deliver a negative thrust
    b. Can be operated in either direction of rotation
    c. Can be installed in either of two ways

69. The term “Constant speed” in reference to propellers means:
    a. A uniform rate of pitch change at all times
    b. The propeller maintains constant r.p.m. in normal conditions
    c. The engine maintains constant r.p.m. in normal operation

70. Which of the following should be checked on a propeller at daily line inspection?
    a. Blade angle settings
    b. Propeller track
    c. Shaft for alignment

71. Which of the following methods would be used to check for cracks in a steel propeller blade?
    a. Magnaflux
    b. Zyglo
    c. Dye check

72. What is meant by a propeller’s “critical range”?
    a. An r.p.m. range during which severe vibrations may occur
    b. The maximum r.p.m. obtainable at high pitch
    c. A propeller’s range of operation from low to high pitch

73. Steel propeller hubs should best be cleaned with:
    a. Emery paper and steel wool
    b. Sand blasting
    c. Soap and water, or gasoline

74. A friction brake is employed on:
    a. All propellers to cause them to stop when feathered
    b. All propellers going into reverse pitch
    c. Electric propellers to stop the pitch changing motor

75. The blade angle (pitch) of a fully feathered propeller is about:
    a. 45°
    b. 20°
    c. 89°

76. If an engine were started on a cold day and ran normally a few revolutions and then stopped, the cause would most likely be:
    a. Booster inoperative
    b. Moisture shorting spark plugs
    c. Weak magneto magnet

77. A battery generator system provides direct current. On installations requiring alternating current from the battery-generator system, it is necessary to have:
    a. A voltage regulator to provide alternating current
    b. A transformer
    c. An inverter

78. A type of battery in which the electrolyte is in a comparatively dry form and which cannot be re-charged is called:
    a. A primary cell
    b. A storage battery
    c. An accumulator

79. Three condensers, each having a capacity of 15 microfarads, are connected in series. The combined capacity of the condensers is:
    a. 5 microfarads
    b. 3.3 microfarads
    c. 45 microfarads

80. A rheostat is a type of:
    a. Transformer
    b. Master switch
    c. Resistor

81. The power actually received by an aircraft propeller is called:
    a. Rated horse power
    b. Brake horse power
    c. Indicated power output

82. A propeller reduction gear is necessary primarily to utilize full engine power and:
    a. Keep engine from over-speeding
    b. Prevent propeller torque
    c. Keep down propeller tip speed

83. A cylindrical tube narrows down at a point when air flows through the tube, there will be:
    a. pressure build-up at this point
    b. depression at this point
    c. temperature rise at this point

84. Bernoulli's theorem is
    a. Momentum law for a gas
    b. Energy law for a gas
    c. Inertial law for a gas

85. When volume of a given mass of gas is doubled at constant temperature its pressure is:
    a. doubled
    b. quadrupled
    c. halved

86. When volume of a given mass of gas is doubled at constant pressure its temperature is:
    a. doubled
    b. halved
    c. quadrupled

87. According to Newton's law the force acting on a body is proportional to:
    a. the rate of action on it
    b. the rate of change of its momentum
    c. Acceleration

88. The approximate governing gas law during the compression stroke of an engine is:
    a. Charles' law
    b. Boyle's law
    c. Momentum law

89. Late ignition timing is defined as:
    a. Number of degrees of crankshaft rotation at which spark occurs A.T.C.
    b. Number of degrees of crankshaft rotation at which spark occurs after manufacturer's specification
    c. Number of degrees of crankshaft rotation at which spark occurs B.T.C.

90. Compression ratio is ratio of:
    a. Volume of cylinder with piston at B.D.C. to volume at T.D.C.
    b. Piston travel on compression stroke to travel on intake stroke
    c. Mixture of fuel to air in combustion chamber

91. A low tension ignition system is:
    a. One having individual coils for each spark plug
    b. A system in which the spark plug voltage is held to only a few hundred volts
    c. Any system designed for use on light aircraft

92. The purpose of the magnet in a magneto is to:
    a. Provide lines of flux
    b. Attracts poles shoes and keep them joined
    c. Stores current to prevent arcing of breaker points

93. If a magneto grounded wire becomes chafed and worn through its insulation until it contacted the engine mount, it would cause:
    a. Failure of magneto to fire plugs
    b. Intermittent missing of engine at all speed
    c. Failure of an engine to stop at switch off

94. Failure of an engine to stop with switch turned off may indicate:
    a. Breaker points sticking
    b. Magneto grounded
    c. Broken ground wire

95. An 18mm spark plug is one that has:
    a. 18 threads per millimeter
    b. A reach of 18 millimeters
    c. A thread diameter of 18 millimeters

96. The four strokes of a four-stroke cycle engine in order of occurrence are:
    a. Power, intake, compression, exhaust
    b. Exhaust, compression, intake, power
    c. Intake, compression, power, exhaust

97. If the wires connecting both primary windings of a dual magneto to the cockpit switch are shorted, the immediate result will be:
    a. Nothing
    b. Failure of engine to continue to operate
    c. Inability to stop the engine

98. Magneto timing consists of degrees before top dead centre on:
    a. Intake stroke
    b. Compression stroke

c. Power stroke

99. Sodium-filled valves are advantageous in that they:

a. Dissipate heat well

b. Cost less

c. Have great strength properties

100. Nitriding is a process given to cylinder barrels to:

a. Anneal them

b. Straighten them

c. **Harden them**